

Progress Test 1.C — Thermo-S1

1.C · Equations of state and state functions

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

Convert the pressure $p = 1013 \text{ hPa}$ into bar. Give the exact result.

$p = 1013 \times 10^2 \text{ Pa} = 1.013 \times 10^5 \text{ Pa}$	and	$1 \text{ bar} = 10^5 \text{ Pa}$
$p = 1.013 \text{ bar}$		

Marking

①
0 1

Marking scheme

① 1 pt 1.013 bar.

Exercise 2. Equation of state and state function — definitions

(5 marks)

Define an equation of state. Write the total differential dV of the state function $V(T, p)$. Give the meaning of the partial derivative $\left(\frac{\partial V}{\partial p}\right)_T$, and its geometric interpretation in a state diagram.

An equation of state is a relation $f(p, V, T) = 0$ (at given n) between the state variables of a system at equilibrium. It characterises the substance and holds only at equilibrium.
$dV = \left(\frac{\partial V}{\partial T}\right)_p dT + \left(\frac{\partial V}{\partial p}\right)_T dp$
$\left(\frac{\partial V}{\partial p}\right)_T$ is the rate of change of V with p at fixed T. It is the slope of the isotherm $V(p)$ in the (p, V) diagram, and it is negative for a gas.

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts Relation between p , V , T (1 pt), valid only at equilibrium and specific to the substance (1 pt).
- ② 2 pts Total differential with both partial derivatives and their fixed variables.
- ③ 1 pt Variation with p at fixed T ; slope of the isotherm.

Exercise 3. State diagrams

(4 marks)

The curve $V(p)$ measured at constant temperature is called:

- ☒ an isotherm ☐ an isobar ☐ an isochore ☐ an adiabat

Why: “iso-therm”: same temperature. At constant p it would be an isobar; at constant V , an isochore.

For a low-density gas, $\left(\frac{\partial V}{\partial p}\right)_T$ is:

- ☐ positive ☒ negative ☐ zero ☐ infinite

Why: at fixed T , compressing the gas (larger p) reduces its volume (Fig. 7 of the notes).

On the curve $V(T)$ at constant p , $\left(\frac{\partial V}{\partial T}\right)_p$ is:

- ☒ the slope of the curve ☐ the area under the curve
☐ the intercept at $T = 0$ ☐ the value of V

Why: a partial derivative is read as the slope of the state diagram in which the other variable is held fixed.

An equation of state is valid:

- ☐ during any transformation ☒ only at thermodynamic equilibrium
☐ only for ideal gases ☐ only at 0°C

Why: “By construction, the equation of state is valid only at thermodynamic equilibrium” (Box 9).

Exercise 4. Partial derivatives of a state function

(4 marks)

A gas obeys the equation of state $V = nRT/p$, which is given here. Compute $\left(\frac{\partial V}{\partial T}\right)_p$ and $\left(\frac{\partial V}{\partial p}\right)_T$ in the state below. Then compute the variation dV when T increases by dT and p increases by dp at the same time. Comment.

$n = 1.0 \text{ mol}$	$R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$	$T = 300 \text{ K}$	$p = 1.0 \times 10^5 \text{ Pa}$
	$dT = 3.0 \text{ K}$	$dp = 1.0 \times 10^3 \text{ Pa}$	

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{nR}{p} = \frac{1.0 \times 8.3}{1.0 \times 10^5} = 8.3 \times 10^{-5} \text{ m}^3 \text{ K}^{-1}$$

$$\left(\frac{\partial V}{\partial p}\right)_T = -\frac{nRT}{p^2} = -\frac{8.3 \times 300}{10^{10}} \approx -2.5 \times 10^{-7} \text{ m}^3 \text{ Pa}^{-1}$$

$$dV = 8.3 \times 10^{-5} \times 3.0 - 2.49 \times 10^{-7} \times 1.0 \times 10^3 = 2.49 \times 10^{-4} - 2.49 \times 10^{-4}$$

$$dV = 0$$

Dividing by V : $\frac{dV}{V} = \frac{dT}{T} - \frac{dp}{p} = 1\% - 1\% = 0$. T and p rise by the same relative amount, so their two effects on V cancel.

Marking

①		
0	1	
②		
0	1	
③		
0	1	
④		
0	1	

Marking scheme

- ① 1 pt $\left(\frac{\partial V}{\partial T}\right)_p = nR/p$ (literal).
- ② 1 pt $\left(\frac{\partial V}{\partial p}\right)_T = -nRT/p^2$ (literal).
- ③ 1 pt Both values, with units $\text{m}^3 \text{ K}^{-1}$ and $\text{m}^3 \text{ Pa}^{-1}$.
- ④ 1 pt $dV = 0$, explained.

Examiner's tip

The unit of a partial derivative is (unit of the numerator)/(unit of the variable), e.g. $\text{m}^3 \text{ K}^{-1}$. Writing it is exactly the “where are the units?” remark of Test 1.

Your score: /14 $\longrightarrow$ mark out of 20 = score $\times$ 20/14 = /20